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Open vehicle routing problem with split deliveries: mathematical formulations and a cutting-plane method

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Abstract

This study addresses the open vehicle routing problem with split deliveries, a variant of the classical vehicle routing problem that allows open routes and partitioned deliveries for customers (i.e., multiple vehicles may serve one customer). This approach may be beneficial for companies interested in reducing their logistics and distribution costs. This study is motivated by a company in the pharmaceutical industry, that seeks to explore the possibility of splitting their deliveries to improve quality indicators related to on-time deliveries and customer satisfaction. Two mixed-integer formulations of the problem are proposed. Additionally, a cutting-plane method is designed to improve the optimization performance. Computational experiments were conducted to validate the performance of the formulations. The second formulation effectiveness was confirmed by providing optimal solutions for instances of up to 30 nodes in a reasonable computational time. The incorporation of the cutting-plane method improves performance with a substantial reduction in the GAP. In the case study, this new approach shows its effectiveness in economic terms by providing savings of up to 20% of the current distribution costs.

Keywords Open vehicle routing · Mixed integer programming · Split delivery · Logistic distribution · Cutting-plane method · Mixed integer formulations

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1 Introduction

The vehicle routing problem (VRP) that considers a fleet of vehicles available to serve a group of customers with known demand has been widely studied in the literature (Marinakis and Migdalas 2007). Each client must be visited by exactly one vehicle, all of which leave from and return to a depot. The objective is to minimize the total distance traveled while satisfying customers' demand (Toth and Vigo 2002). In particular, the deterministic version of the VRP is usually studied, although recent studies have been focusing on the stochastic version of the problem (Rajabi-Bahaabadi et al. 2019). Some variants of the classic VRP consider partitioning of deliveries, so that a customer demand is partitioned in two or more packages and therefore visited by two or more vehicles. This problem is known as the split delivery vehicle routing problem (SDVRP) and has been previously studied in Archetti et al. (2008) and Archetti and Speranza (2008). By partitioning deliveries, significant savings in total costs can be achieved. Another variant less often studied in the literature is the open vehicle routing problem (OVRP). In the OVRP vehicles do not have to return to the depot, but end their routes at the location of some client or node other than the depot. The OVRP procures savings in the total distance traveled (Li et al. 2007) by not forcing vehicles to return to the depot when not necessary.

The first contributions to the SDVRP were made by Dror and Trudeau (1989) and Moshe and Pierre (1990), who showed that savings could be achieved by allowing the partitioning of deliveries. The main articles on the SDVRP were written by Archetti and Speranza (2008), Archetti et al. (2006) and Archetti et al. (2008), who analyzed the maximum possible savings in the partitioning of deliveries and concluded that savings strongly depend on the characteristics of the instances under study. Additionally, they proposed an efficient metaheuristic algorithm to solve this problem. More recently, Ozbaygin et al. (2018) presented new exact approaches for the SDVRP. There are other variants such as the one by Batsyn et al. (2019) wherein vehicles are subjected to site-depended constraints.

A problem less studied at the moment is that of routings that combine these two variants, i.e., an OVRP that allows deliveries to be partitioned, particularly an open vehicle routing problem with split deliveries (OVRPSD). In this paper, the OVRPSD is studied, considering the hiring cost (HC) for each vehicle to be used. To the best of our knowledge, several articles addressed the OVRPSD. Among them, we highlight the work by Xia and Fu (2018a) that presented an algorithm based on tabu search for the OVRP with split deliveries by order. They also presented an improved tabu search algorithm for the OVRP with some particular characteristics in another study (Xia and Fu 2018b). In Ozbaygin et al. (2018), an arc flow formulation for the SDVRP that uses variables with vehicle indices is proposed. The authors also perform computational experiments considering open and closed routes. As for the OVRP, many articles are reported in the literature, we can highlight that by Lalla-Ruiz et al. (2016) who presented a multi-depot open vehicle routing problem. They

introduced a new mathematical formulation that shows its effectiveness over previous work in this area.

In this study, two mathematical models for the OVRPSD are presented, and valid inequalities for a cutting-plane method are developed to improve the model's performance. The computational results show that the proposed mathematical formulations are suitable for this type of problem, and the method with cutting planes obtains very good results when tested over instances of up to 30 nodes.

A real situation faced by a pharmaceutical firm with more than 20 years in the market inspired the problem under study. The company required to reduce its distribution costs by considering split deliveries and some particular constraints. Therefore, the proposed model considers a fleet of homogeneous vehicles (with a given capacity), split deliveries, route distance constraints, and non-split terminal vertices. Non- split terminal vertices, refers that two routes are unable to end at the same client, due to company policy. Further explanation can be found in Sect. 6.

The remainder of the paper is organized as follows. Section 2 describes the problem and the proposed mathematical formulations. Section 3 presents the results of the preliminary experiments with formulations without the cutting planes. Section 4 presents some valid inequalities and also describes the proposed method of cutting planes. Section 5 shows the computational experimentation. The case study is described in Sect. 6. Finally, Sect. 7 describes the relevant findings and proposes possible future lines of research

2 Mathematical formulations for the OVRPSD

In this section, the mathematical formulations developed for the OVRPSD are presented. This problem can be formally defined as follows: Let $G = (V, A)$ be a directed graph, with $V = \{0, 1, 2, \ldots, n\}$ denoting the set of nodes, where 0 refers to the depot and $V^+ = \{1, 2, \ldots, n\}$ to the customer set, and $A = \{(i,j), i \in V, j \in V^+; i \neq j\}$ denoting the set of arcs. There is a distance parameter $t_{ij} > 0$ and a cost $c_{ij} > 0$ associated with each arc $(i,j) \in A$. In particular, the cost c_{ij} can represent time spent, fuel consumption, cost per km, etc. and can be proportionally equivalent to the distance between nodes i and j with appropriate adjustments. The cost matrix is assumed to be symmetric and satisfies the triangle inequality. Each customer $i \in V^+$ has a demand d_i and the depot supplies enough goods to serve all customers. In addition, there are a finite number of vehicles with identical capacity (Q) and hiring cost (H). Finally, there is a limited distance allowed per route (L) and the total capacity of all vehicles is enough to serve all customers. The objective is to minimize the total cost associated with both, the vehicle HC and the total distance traveled to serve all customers.

2.1 First mathematical formulation for the OVRPSD

The first formulation is based on the classical approach for the VRP. Thus, the variables associated with the model are the following: a binary variable x_{ij}^k is equal to 1 when the

arc $(i,j) \in A$ is used by vehicle k for transportation between nodes i and j and 0 otherwise. Let z^k be a binary variable equal to 1 if vehicle k is selected for starting any route and equal to 0 otherwise (as defined by Repoussis et al. (2007)). Additionally, let s_i be a binary variable equal to 1 if node i represents the final node of the route and equal to zero otherwise. Considering this definition, we also assume that the terminal node for a route cannot simultaneously be a final vertex in another path. However, an ending node for one path can be an intermediate node for another path. Let u_{ij}^k be a continuous variable that estimates the cumulative demand of the route after visiting node i by vehicle k. Finally, let q_i^k be a continuous variable that indicates the amount of demand delivered to node i by vehicle k.

Our first formulation is stated as follows:

Minimize:

$$\sum_{k\in K}\sum_{i\in V}\sum_{j\in V^+} c_{ij}x_{ij}^k + \sum_{k\in K} Hz^k \tag{1}$$

subject to:

$$\sum_{k\in K}\sum_{j\in V^+} x_{0j}^k \le |K| \tag{2}$$

$$\sum_{k\in K}\sum_{j\in V^+} x_{0j}^k = \sum_{j\in V^+} s_j \tag{3}$$

$$\sum_{j\in V^+} x_{0j}^k \le z^k \quad \forall k \in K \tag{4}$$

$$\sum_{k\in K}\sum_{i\in Vi\neq j} x_{ij}^k \ge 1 \quad \forall j \in V^+ \tag{5}$$

$$\sum_{i\in Vi\neq j} x_{ij}^k = \sum_{i\in V^+ i\neq j} (x_{ji}^k + s_j) \quad \forall j \in V^+, k \in K \tag{6}$$

$$\sum_{k\in K} q_i^k = d_i \quad \forall i \in V^+ \tag{7}$$

$$q_i^k \le d_i \sum_{j\in Vj\neq i} x_{ji}^k \quad \forall i \in V^+, k \in K \tag{8}$$

$$\sum_{i\in V}\sum_{j\in V^+ j\neq i} t_{ij}x_{ij}^k \le Lz^k \quad \forall k \in K \tag{9}$$

$$u_{0i}^{k} + \sum_{j \in Vj \neq i} u_{ji}^{k} - \sum_{j \in V^{+} j \neq i} u_{ij}^{k} = q_{i}^{k}, \quad \forall i \in V^{+}, k \in K \tag{10}$$

$$u_{ij}^{k} \geq q_{j}^{k} - Q(1 - x_{ij}^{k}) \quad \forall i \in V, \forall j \in V^{+}, k \in K \tag{11}$$

$$u_{ij}^{k} \leq Q x_{ij}^{k} \quad \forall i \in V, \forall j \in V^{+}, k \in K \tag{12}$$

$$x_{ij}^{k} \in \{0, 1\} \quad \forall i \in V, j \in V^{+}, k \in K \tag{13}$$

$$s_{j} \in \{0, 1\} \quad \forall j \in V^{+}, k \in K \tag{14}$$

$$z^{k} \in \{0, 1\} \quad \forall k \in K \tag{15}$$

$$u_{ij}^{k} \geq 0 \quad \forall i \in V, j \in V^{+}, k \in K \tag{16}$$

$$q_{i}^{k} \geq 0 \quad \forall i \in V^{+}, k \in K \tag{17}$$

In this formulation, the objective function (1) models the trade-off between vehicle and routing costs. Constraint (2) ensures that the number of selected vehicles does not exceed the size of the fleet (the maximum number of available vehicles). Constraint (3) ensures that there are the same number of starting and ending nodes. These constraints impose that every single route must depart from the origin node (depot) and end at a customer node (avoiding returning to the depot). The constraint (4) activates the specific vehicles that depart from the depot and serve the routes. Constraint (5) allows that every node j must be served by at least one vehicle. Constraint (6) balances the degree of every node j in terms of the number of entering and leaving arcs. In other words, for a given customer j, the same amount of vehicles that enter must leave. Constraint (7) ensures that the demand of each node i is fulfilled (by at least one vehicle) while constraint (8) assures that the maximum amount of load delivered by any vehicle k that traverse through node i does not exceed the demand of this node, these constraints are adapted from Archetti and Speranza (2008). The family of constraint (9) ensures that, for all of the routes, their respective vehicles do not exceed the maximum allowed travel distance. Constraint (10) corresponds to a generalization of the Miller-Tucker-Zemlin (MTZ) constraints, estimating the cumulative demand on the vehicles and eliminating sub-tours. Constraints (12) and (11) impose the minimum and maximum allowed values for variables u_{ij}^{k}. These constraints are appropriately derived from the study by Nucamendi et al. (2018). Finally, the group of constraints (13–17) establish the nature of the variables.

2.2 Second mathematical formulation for the OVRPSD

For the second proposed formulation, variables u_{ij}^k are simplified as u_i^k but maintain the estimation of the cumulative load for the vehicle k after visiting node i. Additionally, variables z^k can be eliminated since the HC can be included in the departing arc costs (from the depot to any other customer node) using $c_{0i} = c_{0i} + H$. If vehicles have a different hiring cost H_k, a new arc cost $c_{0ik} = c_{0i} + H_k$ associated with each vehicle $k \in K$, can be used.

These substitutions reduce the size of the problem by decreasing the number of variables and constraints.

Considering these changes, the new formulation is as follows:

Minimize:

$$\sum_{k\in K}\sum_{i\in V}\sum_{j\in V^+} c_{ij}x_{ij}^k \tag{18}$$

subject to:

$$\sum_{k\in K}\sum_{j\in V^+} x_{0j}^k = \sum_{j\in V^+} s_j \tag{19}$$

$$\sum_{j\in V^+} x_{0j}^k \le 1 \quad \forall k \in K \tag{20}$$

$$\sum_{i\in V}\sum_{k\in K} x_{ij}^k \ge 1 \quad \forall j \in V^+ \tag{21}$$

$$\sum_{i\in V}\sum_{k\in k} x_{ij}^k = \sum_{i\in V^+}\sum_{k\in k} x_{ji}^k + s_j \quad \forall j \in V^+ \tag{22}$$

$$\sum_{i\in V} x_{ij}^k \ge \sum_{i\in V^+} x_{ji}^k \quad \forall j \in V^+, k \in K \tag{23}$$

$$\sum_{k\in K} q_i^k = d_i \quad \forall i \in V^+ \tag{24}$$

$$d_i \sum_{j\in Vj\neq i} x_{ji}^k - q_i^k \ge 0 \quad \forall i \in V^+, k \in K \tag{25}$$

$$\sum_{j\in Vj\neq i} x_{ji}^k - q_i^k \le 0 \quad \forall i \in V^+, k \in K \tag{26}$$

$$\sum_{i \in V} \sum_{j \in V^+ j \neq i} t_{ij} x_{ij}^k \leq L \quad \forall k \in K \tag{27}$$

$$u_i^k - u_j^k + Q x_{ij}^k \leq Q - q_j^k \quad \forall i \in V, j \in V^+, k \in K \tag{28}$$

$$x_{ij}^k \in \{0,1\} \quad \forall i \in V, j \in V^+, k \in K \tag{29}$$

$$s_j \in \{0,1\} \quad \forall j \in V^+ \tag{30}$$

$$u_i^k \geq 0 \quad \forall i \in V, k \in K \tag{31}$$

$$q_i^k \geq 0 \quad \forall i \in V^+, k \in K \tag{32}$$

To formulate this second model, since the HC of every selected vehicle is included within the cost of the departing arcs, variables z^k are no longer necessary for the objective function. Consequently, this formulation only requires Eq. (20) instead of Eqs. (2) and (4). Similarly, because variables z^k are no longer used, constraint (27) is streamlined to ensure that no vehicle exceeds the maximum allowed travel distance. Similarly, constraint (28) is a simplified version of constraint (10). Finally, constraints (29–32) denote the nature of the variables. We remark that both models present the limitations of not allowing split delivery in the final (last) node on any route.

It is important to note that considering a homogeneous fleet facilitates the elimination of z_k (since all the HCs are the same). For the case of a heterogeneous fleet, the variables z_k must be preserved, as well as Eqs. (2) and (4) should be restored instead of Eq. (20).

3 Preliminary experiments

To verify which one of these formulations provide better results, both models were assessed using a set of six test instances (two considering 20 vertices, two of 25 and two of 30). These instances were randomly generated based on the criteria reported by Archetti et al. (2008) to encourage splitting. For each instance, three different *HC* values were used (0, 50, and 100). The mathematical models were implemented in C++ using Visual Studio and solved using CPLEX 12.6.1, and a time limit of 10,800 s was used. A workstation computer with an Intel Xeon CPU at 3.40 GHz (10 cores) was used. The results are summarized in Tables 1 and 2, where column *HC* shows the hiring cost, *InitialLB* shows the value for objective function of the linear relaxation of the formulation, *UB* shows the upper bound found by the formulation, *GAP* shows the percentage deviation between the upper bound (UB) and the final lower bound (LB)

Table 1 Summary results obtained by the formulation 1

Instance	HC	n	Initial LB	UB	GAP	CPU
G20a	0	20	260.73	296.45	3.81	10,800.00
G20b	0	20	203.04	227.23	3.02	10,800.00
G25a	0	25	258.93	280.31	0.00	646.33
G25b	0	25	232.86	259.52	0.00	9763.91
G30a	0	30	370.05	416.54	2.55	10,800.00
G30b	0	30	240.21	278.45	0.00	499.85
G20a	50	20	547.20	598.91	2.03	10,800.00
G20b	50	20	543.72	581.58	1.91	10,800.00
G25a	50	25	491.39	533.92	0.50	10,800.00
G25b	50	25	472.95	509.89	2.58	10,800.00
G30a	50	30	649.08	716.54	1.41	10,800.00
G30b	50	30	429.09	487.51	0.00	7619.59
G20a	100	20	833.67	900.91	5.40	10,800.00
G20b	100	20	884.40	931.58	1.21	10,800.00
G25a	100	25	723.84	783.92	0.46	10,800.00
G25b	100	25	713.04	759.89	3.36	10,800.00
G30a	100	30	928.12	1016.54	0.96	10,800.00
G30b	100	30	617.24	687.51	0.55	10,800.00
Avg			522.19	570.40	1.65	9429.42

Table 2 Summary results obtained by the formulation 2

Instance	HC	n	Initial LB	UB	GAP	CPU
G20a	0	20	225.14	296.45	0.00	1247.77
G20b	0	20	178.04	227.23	0.00	17.33
G25a	0	25	241.16	280.31	0.00	48.15
G25b	0	25	222.19	259.52	0.00	91.19
G30a	0	30	314.43	419.52	14.08	10,800.00
G30b	0	30	217.92	278.45	0.00	867.34
G20a	50	20	525.14	598.91	0.00	3306.77
G20b	50	20	528.04	581.58	0.00	87.08
G25a	50	25	491.16	533.92	0.00	366.54
G25b	50	25	472.19	509.89	0.00	115.45
G30a	50	30	614.43	716.54	8.01	10,800.00
G30b	50	30	417.92	487.51	2.90	10,800.00
G20a	100	20	825.14	898.91	0.00	5435.90
G20b	100	20	878.04	931.58	0.00	76.59
G25a	100	25	741.16	783.92	0.00	745.96
G25b	100	25	722.19	759.89	0.00	126.73
G30a	100	30	914.43	1020.33	4.98	10,800.00
G30b	100	30	617.92	687.51	1.88	10,800.00
Avg			508.14	570.66	1.76	3696.26

found, and *CPU* shows the CPU time in seconds used by the formulation. Finally, row *Avg* shows the average value for all test instance and the columns mentioned above.

These preliminary results show that for the first formulation the average GAP is 1.65% and the average CPU time is 9, 429.42 s. For the second formulations the average GAP and the average CPU time were 1.76% and 3, 696.26 s, respectively. These preliminary results also show that the linear relaxation of the first formulation obtains tighter LBs than the second formulation. In spite of these tighter LBs, the first formulation only solves 4 of the 18 instances to optimality, while the second formulation finds the optimal solution for 13 of the 18 instances. To verify whether the second formulation performs better than the first one in terms of quality of the solution, we performed a Mann-Whitney test to determine if the GAP produced by the first formulation is statistically greater than that obtained by the second, with a significance level of 0.05. The resulting p-values were 0.032 (not adjusted for ties) and 0.025 (adjusted for ties) supporting the conclusion that there is sufficient evidence to determine that the first formulation yields higher GAP values. This behavior can be explained by the fact that the first formulation improves the LB of linear relaxation because of the variable u_{ji}^{k} and restrictions (3) and (4). However, the structure of this variable increases the size of the problem by increasing the number of variables by $n^2|K| - n|K| + |K|$ and increasing the number of restrictions by $n^2|K| - n + 1$. Therefore, a considerable amount of time is required to solve the subproblems during the branch and bound procedure, increasing the elapsed CPU time.

Considering these preliminary results, it was decided to develop a cutting plane algorithm based on formulation 2 using the known valid inequalities presented in Sect. 4. The use of this algorithm had three goals: i) to improve the LB of the linear relaxation at the root node, ii) to solve more instances to optimality, and iii) to reduce the CPU time.

4 Valid inequalities and the proposed cutting-plane method

To improve the performance of the second formulation, a cutting-plane algorithm is proposed. In this algorithm, the well-known *rounded-capacity* inequalities (Araque et al. 1990) are separated using a max-flow min-cut algorithm. These capacity inequalities are expressed as follows:

$$\sum_{k\in K}\sum_{i\in V\setminus D}\sum_{j\in D} x_{ij}^{k} \geq \left\lceil \frac{d(D)}{Q} \right\rceil \tag{33}$$

The set D contains all vertices connected after vertex i.

The cutting-plane algorithm starts using the objective function (18) along with constraints (20) through (27). Constraint (28) is excluded and the integrality constraints (29)- (30) over the variables are relaxed. The relaxed problem P_r is solved and a fractional solution x_r^* is obtained. Using x_r^* a graph $G^* = (V, A^*)$ where $A^* = (i,j)|\sum_{k\in K} x_{ij}^{k*} > 0$ is created. The capacity in each arc $(i,j) \in A^*$ is defined as $\sum_{k\in K} x_{ij}^{k*}$. To separate the *rounded-capacity* inequalities, a max-flow min-cut algo-

rithm such as the Ford–Fulkerson algorithm is used over graph G^*. The max-flow between the depot and every vertex $j \in V^+$ using graph G^* is computed. If the maximum flow from the depot to any vertex $j \in V^+$ is less than $\left\lceil \frac{d(D)}{Q} \right\rceil$, then the inequality is added to the problem. Figure 1 shows a fractional-solution example. For the example, the set D contains all vertices connected after vertex i ($D = \{h, j, k, l, m\}$) because the maximum flow to any of the vertices in D will be at most 0.5.

When the cutting-plane algorithm is not able to find any new cuts, constraint (28) is added to the model as well as the variable integer constraints. This final model is solved until the maximum CPU time is reached.

The cutting-plane algorithm substantially reduces the average GAP, and in almost all instances, optimality is reached. This can be observed in Table 3.

5 Computational results

This section is devoted to reporting the performance of the developed procedures and discussing the quality of the results obtained in terms of elapsed time and quality of the reported solution. It is divided into two main subsections: (1) description of the instances used to evaluate the behavior of the formulation 2 with cutting planes and (2) detailing the experimental results obtained.

5.1 Test instances

Extensive additional experimentation was conducted over a set of randomly generated instances to validate the performance of the proposed formulation. Three more sets of 10 instances each were generated by following the procedure described in

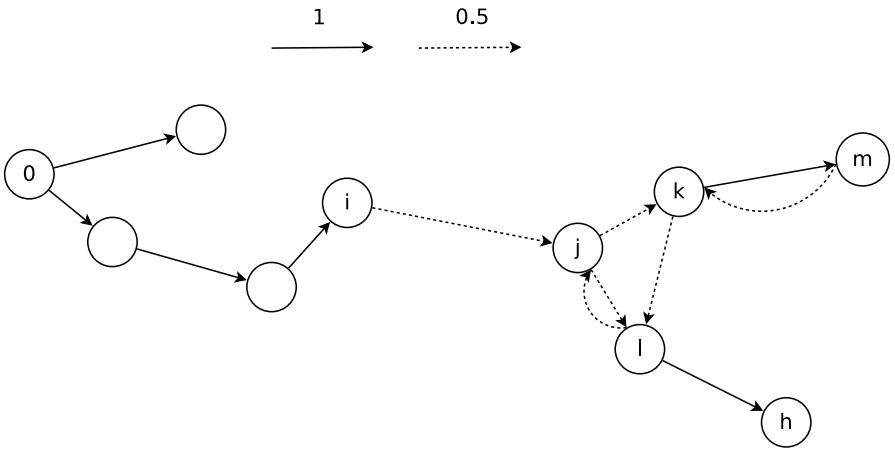


Fig. 1 Fractional solution example

Table 3 Results obtained by cutting plane algorithm with formulation 2 for the preliminary experiments

Instance	HC	n	Initial LB	UB	GAP	CPU
G20a	0	20	279.82	296.45	0.00	357.57
G20b	0	20	223.91	227.23	0.00	6.87
G25a	0	25	275.00	280.31	0.00	8.83
G25b	0	25	250.99	259.52	0.00	62.33
G30a	0	30	376.70	416.54	8.47	10,800.00
G30b	0	30	277.04	278.45	0.00	4.57
G20a	50	20	579.82	598.91	0.00	641.03
G20b	50	20	573.91	581.58	0.00	490.68
G25a	50	25	517.03	533.92	0.00	874.86
G25b	50	25	500.40	509.89	0.00	69.16
G30a	50	30	676.70	716.54	4.38	10,800.00
G30b	50	30	477.30	487.51	0.00	566.74
G20a	100	20	879.82	898.91	0.00	580.28
G20b	100	20	923.91	931.58	0.00	611.16
G25a	100	25	767.13	783.92	0.00	655.22
G25b	100	25	750.40	759.89	0.00	76.68
G30a	100	30	976.70	1016.54	3.21	10,800.00
G30b	100	30	677.30	687.51	0.00	429.85
Avg			554.66	570.28	0.89	2101.99

Sect. 3 and respecting the three values of HC (0, 50 and 100), called **IESn-i** where $n = \{20, 25, 30\}$ and $i = 1 : 10$. All of the instances are available on request.

5.2 Experimental results

This subsection reports the results obtained from the computational experiments considering the cutting-plane algorithm based on formulation 2. All experiments were executed in the same computational equipment used for the preliminary experiments. The mathematical model was implemented in C++ using Visual Studio (using concert technology) and solved using CPLEX 12.6.1 with time limit of 10800 s.

The results are summarized in Table 4 and the detailed results can be found in the Appendix in Tables 7, 8 and 9.

The experimental results presented in Table 4 show that the problem is difficult to solve exactly even for small size instances. For example, only eight of the 10 instances with 20 vertices were solved to optimality independently of the HC. The other two instances reached the maximum CPU time and did not arrive to an optimal solution for any value of HC. It can be observed that the HC has an effect on the number of vehicles used to serve all clients because for six of the 10 instances the number of vehicles was reduced. For the other four, it remained unchanged. For four of the instances that used fewer vehicles, the number of clients with split delivery

Table 4 Analysis of the effect of HC over the number of routes and splitting vehicles

Instance name	HC = 0				HC = 50				HC = 100			
	GAP	CPU time	k	Splits	GAP	CPU time	k	Splits	GAP	CPU time	k	Splits
IES20-1	0.00	357.57	7	2	0.00	641.03	6	2	0.00	580.28	6	2
IES20-2	0.00	6.87	8	2	0.00	490.68	7	3	0.00	611.16	7	3
IES20-3	2.56	10,800.00	6	0	2.38	10,800.00	5	3	1.59	10,800.00	5	3
IES20-4	0.00	3.31	6	1	0.00	459.36	5	4	0.00	331.46	5	4
IES20-5	0.00	209.61	6	2	0.00	499.43	6	2	0.00	1108.42	6	2
IES20-6	0.00	163.45	5	2	0.00	121.17	5	2	0.00	238.35	5	2
IES20-7	0.00	432.07	7	1	0.00	323.21	7	1	0.00	618.17	7	1
IES20-8	0.00	242.34	7	3	0.00	102.20	6	3	0.00	1701.29	6	3
IES20-9	6.55	10,800.00	7	1	3.76	10,800.00	6	3	2.62	10,800.00	6	3
IES20-10	0.00	461.42	6	1	0.00	344.72	6	1	0.00	764.82	6	1
IES25-1	0.00	8.83	6	1	0.00	874.86	5	1	0.00	655.22	5	1
IES25-2	0.00	62.33	6	0	0.00	69.16	5	1	0.00	76.68	5	1
IES25-3	2.92	10,800.00	6	2	1.39	10,800.00	6	2	0.98	10,800.00	6	2
IES25-4	0.00	4968.01	6	0	1.00	10,800.00	5	1	0.90	10,800.00	5	1
IES25-5	4.57	10,800.00	7	2	4.28	10,800.00	6	2	2.95	10,800.00	6	2
IES25-6	0.00	10.88	6	1	0.00	35.82	5	2	0.00	36.60	5	2
IES25-7	0.54	10,800.00	7	0	0.73	10,800.00	6	1	0.72	10,800.00	6	1
IES25-8	0.00	6653.46	6	0	0.54	10,800.00	6	0	0.69	10,800.00	6	0
IES25-9	4.60	10,800.00	7	1	4.10	10,800.00	6	2	2.91	10,800.00	6	1
IES25-10	0.00	41.36	6	1	0.00	207.04	5	1	0.00	138.35	5	1
IES30-1	8.47	10,800.00	6	1	4.38	10,800.00	6	1	3.21	10,800.00	6	1
IES30-2	0.00	4.57	5	0	0.00	566.74	4	0	0.00	429.85	4	0
IES30-3	3.04	10,800.00	5	2	1.62	10,800.00	5	2	1.27	10,800.00	5	2
IES30-4	0.91	10,800.00	6	1	0.73	10,800.00	6	1	0.42	10,800.00	6	1
IES30-5	0.00	5015.21	6	2	0.00	1477.68	5	2	0.00	1436.65	5	2
IES30-6	0.98	10,800.00	5	1	0.00	2277.88	5	1	0.00	1050.11	5	1
IES30-7	4.22	10,800.00	7	1	3.42	10,800.00	6	1	2.17	10,800.00	6	1
IES30-8	2.17	10,800.00	7	1	1.45	10,800.00	6	2	0.69	10,800.00	6	2
IES30-9	1.74	10,800.00	7	1	2.56	10,800.00	6	3	1.75	10,800.00	6	3
IES30-10	6.22	10,800.00	5	2	3.71	10,800.00	5	2	2.39	10,800.00	5	2

increased. Additionally, it is important to mention that, in this group, the CPU time increased when the HC was > 0.

Similar results are observed for the other groups of instances with 25 and 30 vertices. For the 25 vertex group, the number of instances solved to optimality was four independently of the HC. Two instances were solved to optimality with HC=0 and for the other four, a non optimal solution was found independently of the HC. In eight of the 10 instances, the size of the fleet was reduced and, in

five of them, the number of customers with split delivery increased. Again, the elapsed CPU time increased when HC was > 0.

For two of the instances with 30 vertices, an optimal solution was found independently of the HC. The average GAP is higher for the group of 30-vertex instances than for the others. This can be explained as a result of the increasing the size of the instance, in terms of the number of customers. In this group, five instances decreased the number of required vehicles as the HC increased. Of this five, three increased the number of customers with split delivery.

The most difficult instance to solve was IES30-1 because it reported the largest GAP for all values of HC (8.47%, 4.38%, 3.21%). For almost all instances, the GAP was reduced or remained unchanged as the HC increased (except for instances IES25-7 and IES30-9 in which the GAP was larger for HC = 50).

In general, as the HC increases, the fleet size is reduced and the number of split deliveries increases. This indicates that considering the HC could lead to a great benefit if it is included at the beginning of the route-design process.

6 Case study (CE)

As mentioned before, the proposed problem was inspired in a real situation faced by a pharmaceutical firm with more than 20 years in the market. The company has a distribution center located in Guadalajara, Mexico and ships products to 25 small pharmacies (customers) distributed around the metropolitan area. Currently, the firm contracts a small homogeneous fleet of vehicles from third-party companies and pays a fee for each vehicle. To facilitate the management of deliveries, the company established a policy of delivering the entire demand (in pallets) of a customer using only one vehicle. However, sometimes these pharmacies order large quantities of products that require the use of more than 50% of the total capacity of one truck, which makes it challenging to satisfy the entire demand for two (or more) customers using the same vehicle. The last increases the operating costs and generates delays in deliveries, negatively affecting indicators related to the productivity of the firm, such as on-time deliveries and customer satisfaction. To obtain competitive prices, the company anticipates the contracting of an estimated number of vehicles based on predicted demand and number of customers to serve. According to historical data, 13 to 20 customers are daily served, and the estimated demand ranges between 8 and 75 pallets per customer. A maximum travel distance (in km) is established per route. Thus, the company decided to explore the possibility of splitting the deliveries to minimize the total cost, including the hiring and travelling costs, required to serve all customers. The expected benefits of this approach can be reflected as monetary savings as well as by improving internal key performance indicators.

An important aspect to highlight is that the company has a particular policy of ending every single route at a different node. In other words, one node cannot simultaneously be an ending node for two routes. The last, because when considering split deliveries, one part of the total demand must be delivered early

(as soon as possible). They also take into account that the distribution center is located at the center of the city, and the closest pharmacies have a higher demand than peripheral ones. However, a final node of a specific route can be an intermediate node in another route at the same time.

6.1 CE instances

As mentioned above, the company's actual delivery system first estimates the number of vehicles needed to attend to all the customers without considering split deliveries. Then the method for assigning customers to vehicles is performed, followed by the mechanism of designing the route for each vehicle separately. As a result, there are some vehicles with high values of unutilized capacity. Thus, the logistics department is interested in exploring the possibility of partitioning deliveries with the aim of minimizing the number of required vehicles without significantly increasing the traveling cost per vehicle.

The validation of the proposed model is conducted over a set of CE instances generated using real information provided by the pharmaceutical company. These instances represent the number of customers and demands to deliver in a typical work day. To generate them, the locations of the customers (and the potential locations of the depot) are considered. The Euclidean distances between nodes are considered. In total, five instances were created to simulate a labor week. For each instance, two different HC were considered. To preserve the confidentiality of the data, some information is disguised, the number of customers ranges from 13 to 30, the estimated demand in pallets ranges between 8 and 24 per customer.

6.2 Experimental results for the case study

In this section, the experimental results and relevant insights for the case study are reported. Given the preliminary results, the cutting plane algorithm was used on formulation two to conduct the experiments. Following the interest of the company, we evaluate the impact of the HC over the number of vehicles used and the decision to split specific deliveries. A comparison between the minimum and maximum values of HC (0 and 100) was conducted. Finally, a comparison with non-split deliveries

Table 5 Case study results

Instance name	HC = 0					HC = 100				
	n	GAP	CPU time	k	Splits	GAP	CPU time	k	Splits	% CS
CE1	21	0.00	25.97	5	0	0.00	128.67	4	2	13.65
CE2	13	0.00	0.67	4	0	0.00	1.28	3	2	21.76
CE3	21	0.00	3.88	4	2	0.00	6.12	4	2	0.00
CE4	14	0.00	0.31	5	0	0.00	22.94	4	1	15.36
CE5	21	0.00	37.01	6	0	0.00	134.85	5	3	12.04

solutions was performed to evaluate whether a split approach would provide an economic benefit. The second model, presented in Sect. 2.1, was modified to obtain solutions without split deliveries. Constraints (24), (25) and (26), were removed and the term q_j^k is replaced with d_j on constraints (28). Table 5 summarizes the results (detailed results are displayed in Table 10 of section 7).

In Table 5, column 1 and column 2 indicate the name and size of the instance, respectively. Columns 3 and 7 display the *GAP* reported after solving the scenario or reaching the maximum allowed CPU time. Columns 4 and 8 report the CPU time spent by the proposed method. Columns 5 and 9 indicate the number of vehicles selected while columns 6 and 10 show if the best feasible solution found requires split deliveries. Finally, column 11 shows the cost savings percentage (*%CS*), between the optimal solution allowing split deliveries and the optimal solution not allowing split deliveries considering $HC = 100$.

The cutting plane method solves all CE instances to optimality. For the solved instances, the maximum elapsed CPU time was 135 s. This confirms the effectiveness of the proposed method for solving real-scenarios with a substantial reduction in the time required to provide the optimal value. It can also be observed that, in most cases (except instance CE3), the number of active vehicles is reduced when the HC increases. As a result, the number of customers having a split delivery increases as well. Additionally, it is worth to mention that the model considering the HC produces savings of up to 21.76% of the total cost. These cost reductions increase the distance travelled by vehicles by a maximum of 8.71%. In terms of the total cost, this represents a maximum increase of 2.23%.

Table 6 compares the company's current costs with the SDOVRP solutions cost. In this table, columns 3 and 6 present the objective function value for solutions provided by the company and the cutting plane algorithm based on formulation 2, respectively, and column 7 presents the obtained cost savings percentage. The obtained results show that the use of split deliveries and exact methods could help the company to reduce its distribution cost. At this moment, the company is evaluating the purchase of an optimization software license to make it possible the implementation of the proposed cutting plane method in practice.

Table 6 Comparison with company's current costs

Instance	Company		SDOVRP			
	k	OF	k	Splits	OF	% CS
CE1	6	693.28	4	2	554.27	25.08
CE2	4	485.32	3	2	353.53	37.28
CE3	6	718.51	4	2	534.83	34.34
CE4	6	743.62	4	1	538.07	38.20
CE5	6	764.83	5	3	665.85	14.87

7 Conclusions and future work

In this paper, the OVRPSD was addressed. Two mathematical formulations were proposed for solving this problem, and a cutting-plane method was developed to improve the performance in both the quality of the solutions and the elapsed CPU time. The relevant contributions of this paper are as follows: (1) development of two mixed-integer formulations capable of solving SDOVRP instances of up to 30 customers, (2) development of an efficient cutting plane method that substantially improves their performance and (3) validation of the benefits of the proposed method over a real-life case study.

From the results, it can be observed that the second formulation outperforms the first one. The improvement in CPU time is as substantial as 39%. Additionally, the implementation of a cutting-plane algorithm for the second formulation allows finding optimal solutions for almost all the test instances and reduces the overall average CPU time.

For the case study, it was shown that the proposed method could provide benefits for the company in both aspects: (i) producing savings up to 20% in the total cost when compared to the traditional VRP currently implemented and (ii) by obtaining optimal solutions in a significantly shorter time. Additionally, the increment in travelling costs represented at most 2.23% of the total cost, which supports the fact that partitioning deliveries, produces significant savings in terms of the number of vehicles used and, it can also allow the company to maintain a right level of customer satisfaction.

Future lines of research include considering more realistic scenarios such as multidepot scenarios, dynamic environments and stochastic service times. The incorporation of time windows is also a research line that can be explored.

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Appendix: Results of computational experiments

The summary results for the benchmark problems are presented in Tables 7, 8 and 9. In such tables, columns 1, 2, and 3 denote the name of the instance, the HC, and the size of the fleet reported. Column 4 shows the initial Lower Bound (LB), column 5 the Best Integer solution found, column 6 shows the final Lower Bound (LB), and column 7 the GAP in percentage. Column 8 reports the total elapsed CPU time (including the cutting plane phase) to obtain the integer solution. Finally, the last column indicates the number of split deliveries.

Table 7 Results obtained by the cutting plane and formulation 2 with HC=0

Instance name	HC	k	Initial LB	Best integer	LB	GAP	CPU time (s)	Number of splits
IES20-1	0	7	279.82	296.45	296.45	0.00	357.57	2
IES20-2	0	8	223.91	227.23	227.23	0.00	6.87	2
IES20-3	0	6	289.88	318.00	310.06	2.56	10,800.00	0
IES20-4	0	6	253.18	254.68	254.68	0.00	3.31	1
IES20-5	0	6	264.95	279.29	279.29	0.00	209.61	2
IES20-6	0	5	273.70	288.10	288.10	0.00	163.45	2
IES20-7	0	7	280.38	291.46	291.46	0.00	432.07	1
IES20-8	0	7	222.83	237.10	237.10	0.00	242.34	3
IES20-9	0	7	346.11	381.54	358.08	6.55	10,800.00	1
IES20-10	0	6	297.52	316.35	316.35	0.00	461.42	1
IES25-1	0	6	275.00	280.31	280.31	0.00	8.83	1
IES25-2	0	6	250.99	259.52	259.52	0.00	62.33	0
IES25-3	0	6	331.61	350.97	341.01	2.92	10,800.00	2
IES25-4	0	6	263.79	281.40	281.40	0.00	4968.01	0
IES25-5	0	7	300.02	327.67	313.34	4.57	10,800.00	2
IES25-6	0	6	263.60	268.22	268.22	0.00	10.88	1
IES25-7	0	7	292.90	313.46	311.78	0.54	10,800.00	0
IES25-8	0	6	278.51	292.41	292.41	0.00	6653.46	0
IES25-9	0	7	304.27	329.84	315.33	4.60	10,800.00	1
IES25-10	0	6	270.90	280.10	280.10	0.00	41.36	1
IES30-1	0	6	376.70	416.54	384.01	8.47	10,800.00	1
IES30-2	0	5	277.04	278.45	278.45	0.00	4.57	0
IES30-3	0	5	321.14	346.94	336.69	3.04	10,800.00	2
IES30-4	0	6	279.26	292.28	289.64	0.91	10,800.00	1
IES30-5	0	6	350.28	362.73	362.73	0.00	5015.21	2
IES30-6	0	5	306.53	321.75	318.62	0.98	10,800.00	1
IES30-7	0	7	322.60	344.38	330.45	4.22	10,800.00	1
IES30-8	0	7	236.17	252.88	247.51	2.17	10,800.00	1
IES30-9	0	7	317.03	339.36	333.55	1.74	10,800.00	1
IES30-10	0	5	299.86	335.76	316.10	6.22	10,800.00	2
Average						1.64	5561.07	

Table 8 Results obtained by the cutting plane and formulation 2 with HC=50

Instance name	HC	k	Initial LB	Best integer	LB	GAP	CPU time (s)	Number of splits
IES20-1	50	6	579.82	598.91	598.91	0.00	641.03	2
IES20-2	50	7	573.91	581.58	581.58	0.00	490.68	3
IES20-3	50	5	539.88	571.41	558.12	2.38	10,800.00	3
IES20-4	50	5	504.30	522.01	522.01	0.00	459.36	4
IES20-5	50	6	564.06	579.29	579.29	0.00	499.43	2
IES20-6	50	5	524.03	538.10	538.10	0.00	121.17	2
IES20-7	50	7	630.38	641.46	641.46	0.00	323.21	1
IES20-8	50	6	522.83	537.10	537.10	0.00	102.20	3
IES20-9	50	6	646.11	683.89	659.13	3.76	10,800.00	3
IES20-10	50	6	597.52	616.35	616.35	0.00	344.72	1
IES25-1	50	5	517.03	533.92	533.92	0.00	874.86	1
IES25-2	50	5	500.40	509.89	509.89	0.00	69.16	1
IES25-3	50	6	633.87	650.97	642.04	1.39	10,800.00	2
IES25-4	50	5	513.79	534.82	529.54	1.00	10,800.00	1
IES25-5	50	6	600.02	638.79	612.60	4.28	10,800.00	2
IES25-6	50	5	514.30	521.19	521.19	0.00	35.82	2
IES25-7	50	6	593.35	615.37	610.90	0.73	10,800.00	1
IES25-8	50	6	578.51	592.41	589.22	0.54	10,800.00	0
IES25-9	50	6	604.27	639.49	614.28	4.10	10,800.00	2
IES25-10	50	5	518.13	531.92	531.92	0.00	207.04	1
IES30-1	50	6	676.70	716.54	686.49	4.38	10,800.00	1
IES30-2	50	4	477.30	487.51	487.51	0.00	566.74	0
IES30-3	50	5	571.14	596.94	587.42	1.62	10,800.00	2
IES30-4	50	6	579.26	592.28	588.01	0.73	10,800.00	1
IES30-5	50	5	600.12	613.21	613.21	0.00	1477.68	2
IES30-6	50	5	556.91	571.75	571.75	0.00	2277.88	1
IES30-7	50	6	622.60	653.45	631.81	3.42	10,800.00	1
IES30-8	50	6	536.17	554.23	546.30	1.45	10,800.00	2
IES30-9	50	6	617.03	647.53	631.35	2.56	10,800.00	3
IES30-10	50	5	550.05	585.76	564.80	3.71	10,800.00	2
Average						1.20	5683.03	

Table 9 Results obtained by the cutting plane and formulation 2 with HC=100

Instance name	HC	k	Initial LB	Best integer	LB	GAP	CPU time (s)	Number of splits
IES20-1	100	6	879.82	898.91	898.91	0.00	580.28	2
IES20-2	100	6	923.91	931.58	931.58	0.00	611.16	3
IES20-3	100	5	789.88	821.41	808.53	1.59	10800.00	3
IES20-4	100	5	754.30	772.01	772.01	0.00	331.46	4
IES20-5	100	7	865.69	879.29	879.29	0.00	1108.42	2
IES20-6	100	5	774.03	788.10	788.10	0.00	238.35	2
IES20-7	100	7	980.38	991.46	991.46	0.00	618.17	1
IES20-8	100	6	822.83	837.10	837.10	0.00	1701.29	3
IES20-9	100	6	946.11	983.89	958.74	2.62	10800.00	3
IES20-10	100	6	897.52	916.35	916.35	0.00	764.82	1
IES25-1	100	6	767.13	783.92	783.92	0.00	655.22	1
IES25-2	100	6	750.40	759.89	759.89	0.00	76.68	1
IES25-3	100	5	933.87	950.97	941.72	0.98	10800.00	2
IES25-4	100	5	763.79	784.82	777.78	0.90	10800.00	1
IES25-5	100	6	900.02	938.79	911.87	2.95	10800.00	2
IES25-6	100	5	764.30	771.19	771.19	0.00	36.60	2
IES25-7	100	6	893.35	915.37	908.86	0.72	10800.00	1
IES25-8	100	6	878.51	892.41	886.31	0.69	10800.00	0
IES25-9	100	5	904.27	941.55	914.95	2.91	10800.00	1
IES25-10	100	5	768.13	781.92	781.92	0.00	138.35	1
IES30-1	100	5	976.70	1016.54	984.93	3.21	10800.00	1
IES30-2	100	6	677.30	687.51	687.51	0.00	429.85	0
IES30-3	100	5	821.14	846.94	836.35	1.27	10800.00	2
IES30-4	100	6	879.26	892.28	888.55	0.42	10800.00	1
IES30-5	100	5	850.12	863.21	863.21	0.00	1436.65	2
IES30-6	100	6	806.91	821.75	821.75	0.00	1050.11	1
IES30-7	100	6	922.60	953.45	933.24	2.17	10800.00	1
IES30-8	100	6	838.31	854.23	848.37	0.69	10800.00	2
IES30-9	100	4	917.03	947.53	931.24	1.75	10800.00	3
IES30-10	100	5	800.05	835.76	816.25	2.39	10800.00	2
Average						0.84	5725.91	

Table 10 Detailed results for the case study

Instance name	HC	k	Non-split Int	Split Int	GAP	CPU time (s)	Number of splits
CE1	0	5	141.91	141.91	0.00	25.97	0
CE2	0	4	51.90	51.90	0.00	0.67	0
CE3	0	4	139.39	134.88	0.00	3.88	2
CE4	0	5	135.75	135.75	0.00	0.31	0
CE5	0	6	156.98	156.98	0.00	37.01	0
CE1	100	4	641.91	554.27	0.00	128.67	2
CE2	100	3	451.90	353.53	0.00	1.28	2
CE3	100	4	539.39	534.83	0.00	6.12	2
CE4	100	4	635.75	538.07	0.00	22.94	1
CE5	100	5	756.98	665.82	0.00	134.85	3

The complete results for the case study are presented in table 10. Columns 1, 2 and 3 denote the name of the instance, the HC and the size of the fleet reported. Column 4 shows the optimal solution not allowing split deliveries, whereas column 5 indicates the optimal solution allowing split deliveries. Column 6 reports the GAP in percentage. Finally, columns 7 and 8 report the CPU time required by the formulation to obtain the solution and the number of split deliveries, respectively.

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